VFX 705 - 02

PROJECT 03: Newton's Cradle

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Process

cleanup_scene()

build_sheep(posX, posY, posZ, size, name, reverse, single)
  build body()
  build_head()
  build_hair()
  build_legs()
  build_board(body_group)
  build_arms()

build_frame(x, y, z, size, mirror_offset_z)
build_all_grass(size_mult)

make_noise_shader()
make_wood_shader()
make_metal_shader()

make_light(index, radius, exposure)
setCamera()
setRenderSettings()
renderScene()
main()

Breakdown

①

Bounding box

The bounding box is used to measure the exact size and position of both the sheep body and the board in world space. By calculating their minimum and maximum coordinates, the code determines width and center points, allowing the board to be scaled proportionally and aligned precisely under the body automatically.

```
PAD = 0.8

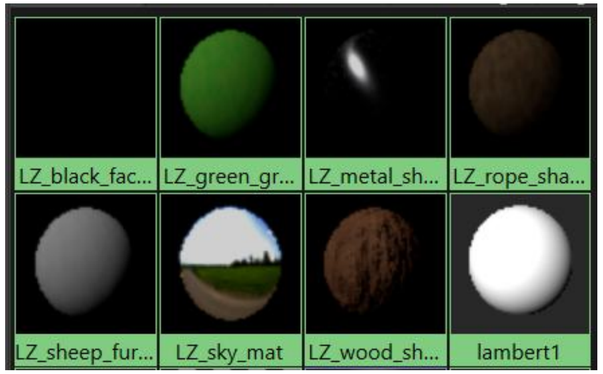
# body bbox
bb = cmds.exactWorldBoundingBox(body_group)
sx_min, sy_min, sz_min, sx_max, sy_max, sz_max = bb
s_cx = (sx_min + sx_max) * 0.5
s_cz = (sz_min + sz_max) * 0.5
s_wx = (sx_max - sx_min)

# board bbox
bb2 = cmds.exactWorldBoundingBox(board)
bx_min, by_min, bz_min, bx_max, by_max, bz_max = bb2
b_cx = (bx_min + bx_max) * 0.5
b_cz = (bz_min + bz_max) * 0.5
b_wx = (bx_max - bx_min)
```

②

Shader

In my code, shaders are built procedurally using aiStandardSurface, combined with aiNoise and ramp nodes to generate textures without image maps.



③

KeyTangent

I used keytangent command to make the camera move more smooth.

```
# Make motion smooth
# itt="auto"  in tangent type (how the motion enters a keyframe)
# ott="auto"  out tangent type (how the motion leaves a keyframe)
cmds.keyTangent(camera, at=["translateX","translateY","translateZ",
"rotateX","rotateY","rotateZ"], itt="auto", ott="auto")
```

Breakdown

④

build_sheep()

```
def build_sheep(posX=0.0, posY=0.0, posZ=0.0, size=1.0, name="sheep",reverse = False, single = False):
```

In the build_sheep() function, I added two additional parameters to control the behavior of the generated sheep:

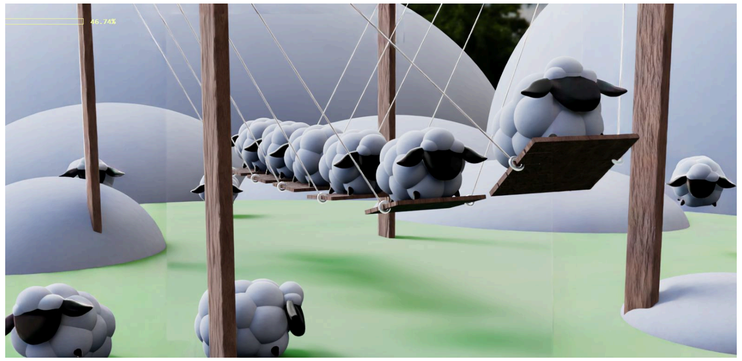
reverse: This parameter controls the facing direction of the sheep. When set to True, the sheep is mirrored by scaling negatively along one axis, effectively flipping its orientation. When set to False, the sheep keeps its default facing direction.

single: This parameter controls whether the sheep includes the swing (arms and board structure). When single=True, the function builds only the basic sheep body parts without the swing components. When single=False, it builds the complete sheep including the swing setup.

⑤

more moving sheep

I gave more sheep expression to make them move



User guide

① Frames

In my code, the total frame count n is defined at the beginning of the script, so it can be easily adjusted before running the animation . The expressions use sin(2 * PI * frame / n), which means the animation always completes exactly one full sine cycle over n frames. As long as n is greater than 1, the motion will return to the same value at the end frame, creating a seamless loop. Changing n only affects the speed of the animation, not whether it loops.

② Hdri used

I used hdri for the sky sphere. User can change add other hdris to change the sky.

Future Improvement

If I had more time to further develop this project, I would extend the system to allow the user to input the number of sheep dynamically. Instead of manually creating each sheep instance, the script would use a loop to automatically generate any number of sheep based on a user-defined parameter. The positioning, spacing, and swing direction would be calculated procedurally so the scene layout remains balanced regardless of quantity. Additionally, I would modify the animation system so the frame range and oscillation timing automatically adapt to the number of sheep, ensuring proper phase offsets and maintaining a seamless looping motion no matter how many sheep are generated.

Updated Version

① A comment header was added at the top of the code to clearly identify the assignment, author, and creation date.

② The value of π is now taken directly from the Python math module (math.pi) instead of being manually defined.

③ All major scene elements are now grouped under a single master group (LZ_MASTER_GRP).

④ The camera setup was improved. The script now explicitly disables renderability for all cameras and enables renderability only for the intended render camera. In addition, the render sequence setting was adjusted so that the scene renders from a single camera instead of attempting to render from all cameras.